

Werner Bisschoff

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Cape Town, Western Cape, South Africa

Embedded Systems & Real-Time Software Engineer

Professional Summary

Rigorous Systems Engineer with 5+ years of experience engineering real-time, event-driven software architectures (C/C++, FreeRTOS) and secure, deterministic communication protocols. Specializes in designing robust firmware, high-performance IPC mechanisms, and hardware-in-the-loop simulation frameworks. Governed by strict Spec-Driven Development (SDD) methodologies and automated validation pipelines to eliminate runtime faults in production environments.

Experience

Founder and Host Cape Town	Divergent Tabletop Jul 2025
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- Engineered and deployed a distributed knowledge-management infrastructure
- Integrated a CRDT synchronization layer (y_ex) to handle real-time modifications; implemented an actor-based buffering model that reduced state-sync latency to sub-50ms thresholds.
- Designed a dynamic BEAM supervision tree with a :one_for_one strategy to isolate concurrent page editing instances, ensuring 99.9% fault isolation.
- Implemented a threshold-based buffer flush and scheduled heap hibernation (hibernate_after), reducing process memory footprints by 60%.
- Applied Monotropic focus paradigms and Universal Design principles to minimize platform cognitive complexity.

Full-Stack Software Engineer Cape Town	FARO Africa Aug 2024 – Nov 2025
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- Built mobile application in Expo, including NFC (ISO 14443-4 APDUs) for e-paper price tags and card operations → enabled real-time price updates with fewer tagging errors
- Introduced LLM-assisted development workflows → improved debugging speed and code review throughput

Embedded Software Engineer Remote Work	Ingenics Digital GmbH (through ViVa Outsourcing) Mar 2023 – May 2024
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- Designed an event-driven finite state machine for an I2C-based embedded system using C++ and FreeRTOS → created maintainable in-house architecture leading to fewer bugs and quicker development
- Developed ESP32 applications using C/C++ and ESP-IDF
- Integrated a configurable low-energy BLE stack for device communication
- Developed an active object within the QP Real-Time Embedded Framework (with QSPY) to simulate device behaviour
- Integrated a configurable BLE stack for device communication and implemented FOTA firmware updates over BLE

Software Developer
Century City, Cape Town

UMAN Technologies
Mar 2021 – Dec 2022

- Implemented RPC services using SOME/IP protocol and reduced performance bottlenecks with perf.
- Built IPC/RPC layers in C++ and Python using Cap'n Proto; designed node trees exposing process variables via IPC interface.

Junior Lecturer
Potchefstroom

North-West University
Feb 2020 – Dec 2020

- Lectured Python and C++ programming for Introduction to Programming for first year IT students.

Education

B.Eng. Computer and Electronic Engineering
Potchefstroom

North-West University
2020

- Focus on embedded systems, software engineering, and electronic design
- Developed NFC payment protocol emulator (Kotlin/Android), STM32 firmware (C), and PID motor controller (Arduino)

Skills

Primary: C++, FreeRTOS, ESP32/ESP-IDF, NFC APDU (ISO 14443-4)

Secondary: NimBLE, UART/SPI/I2C, TinyFrame, Hardware-in-the-Loop (HIL) Testing

Foundational: Assembly, Digital Logic, System Modeling

Other: ERPNext (Integration-focused), Expo Mobile

AI Policy

I treat AI as an agentic partner within a strict engineering framework, moving beyond “vibe-coding” to structured, audit-ready workflows:

Spec-Driven Development (SDD): Prioritize context-heavy specifications to ensure alignment; AI-generated outputs are treated as hypotheses validated by human-led TDD.

Data Security: Enforce mandatory sandboxing for all agentic projects; strictly prohibit credential access and utilize tool-based retrieval for scoped, anonymized data only.

Engineering Rigour: All AI-assisted output undergoes rigorous human-in-the-loop review to ensure it meets enterprise quality and compliance benchmarks.

Privacy by Design: AI workflows are architected to avoid direct database access; sensitive data remains behind scoped, tool-mediated retrieval with no standing credentials exposed to agents.

Job Target

Seeking a mid to senior software engineering role (remote preferred, hybrid max. 2 days/week on-site in Cape Town) with emphasis on embedded systems, backend services, or AI-assisted development workflows. Prefers small, focused teams with clear goals and engineering culture. Open to permanent or contract (3+ months) arrangements.